SAURAV KUMAR

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PROFESSIONAL SUMMARY

Data Science & AI student (B.Tech, 2028) specializing in production-minded Machine Learning and Generative AI. Engineers end-to-end systems, including RAG pipelines, LangGraph multi-agent workflows, MCP tool servers, graph-based anomaly detection and explainable (SHAP) ML APIs, and owns the full lifecycle from EDA and feature engineering to training, MLflow experiment tracking, FastAPI serving, Docker and dashboards. Pairs strong Python, SQL and MLOps skills with a focus on explainable AI that turns data into decisions.

SKILLS

Core Competencies: Machine Learning, Deep Learning, NLP, Generative & Agentic AI, RAG, Data Analysis & Visualization, Model Deployment
Languages: Python, SQL, C++, HTML
ML / AI: LLMs, Multi-Agent Systems, Tool Calling, MCP, Anomaly Detection, Explainable AI (SHAP)
Frameworks & Libraries: PyTorch, TensorFlow, LangChain, LangGraph, LangSmith, FastAPI, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Pydantic, SQLAlchemy
Databases & Big Data: PostgreSQL, MongoDB, MySQL, SQLite, Big Data Querying
Cloud & MLOps: AWS, Kubernetes, Docker, MLflow, Prometheus, Git/GitHub
Tools: Streamlit, Jupyter, Power BI, Excel, VS Code, Colab

EXPERIENCE

Data Science Intern (Virtual) | EduSkills Foundation, with AICTE & Siemens | Certificate **April 2026 June 2026**

- Completed a 10-week program in data engineering, machine learning and applied AI, applying Python, Pandas and NumPy to hands-on data problems.
- Developed and evaluated ML models through data preprocessing, feature engineering and rigorous model evaluation to improve prediction accuracy.

PROJECTS

Clinical Risk & Patient Outcome Intelligence System | *ML / MLOps / Agentic AI* GitHub

- Engineered an end-to-end 30-day readmission predictor on a 5,000-record synthetic dataset (30 features) with data validation and feature engineering, and integrated an LLM agent that calls the trained model through tools (predict, explain, high-risk patients, factors).
- Benchmarked Logistic Regression, Random Forest and Gradient Boosting, auto-selecting the best by ROC-AUC with MLflow tracking; added SHAP explanations and deployed an API-key-secured FastAPI service with audit logs, Prometheus metrics, a Plotly dashboard, Docker Compose and automated tests.
  Tech Stack: Python, Scikit-learn, SHAP, MLflow, FastAPI, SQLAlchemy, PostgreSQL, Groq, Streamlit, Plotly, Prometheus, Docker, Pytest

OpsPilot AI Incident Intelligence Copilot | *Anomaly Detection / RAG / Agent* GitHub

- Developed an incident copilot that detects unusual service metrics with Isolation Forest, retrieves runbooks via TF-IDF/cosine similarity, and orchestrates a tool-using agent (optional Groq LLM) to recommend read-only investigation steps.
- Shipped a REST API (anomalies, investigate, audit endpoints) with request validation, a Streamlit dashboard, SQLite audit logging, Pytest tests and a no-LLM fallback so it runs without an API key.
  Tech Stack: Python, Scikit-learn (Isolation Forest), TF-IDF, FastAPI, Streamlit, SQLite, Groq API, Pytest

Autonomous Cyber Defense System | *Graph ML / Anomaly Detection* GitHub

- Simulated a defensive ML environment by synthesizing network telemetry and modeling traffic as graphs with node- and edge-level features to separate normal from malicious behavior.
- Trained a self-supervised graph autoencoder that flags anomalies by reconstruction error, and implemented a reinforcement-learning policy that selects monitor, rate-limit, isolate or block actions by threat severity.
  Tech Stack: Python, Graph Neural Networks, Autoencoders, Reinforcement Learning, Streamlit

AI Medical Chatbot | *GenAI / RAG* GitHub

- Built a context-aware medical chatbot with LangChain and Hugging Face LLMs, leveraging semantic chunking, embedding retrieval and conversational memory for multi-turn queries.
- Architected a RAG pipeline with FAISS search, cosine-similarity ranking and custom prompts to ground answers in medical knowledge and reduce hallucinations.
  Tech Stack: Python, LangChain, FAISS, Hugging Face, Vector Embeddings, Prompt Engineering

Expense Tracker MCP Server | *LLM Tool Integration* GitHub

- Developed a FastMCP server that lets an AI assistant manage expenses in plain language, exposing tools to add, list by date range and summarize by category, plus a live-editable categories resource.
- Implemented SQLite storage with parameterized queries and auto schema creation, giving LLM clients reliable, structured access to real data.
  Tech Stack: Python, FastMCP, Model Context Protocol (MCP), SQLite, uv

CERTIFICATIONS

Generative AI and Large Language Models Neural Networks and Deep Learning Machine Learning Professional Certification Link
Machine Learning with Python Introduction to Artificial Intelligence Power BI for Data Visualization

EDUCATION

Bachelor of Technology (B.Tech) | **MIT ADT University**, Loni Kalbhor, Pune **2024 2028**